

Quiz 5: Logic Preliminaries

You will have 2-3 minutes to answer this on your own on your sheet (numbers 8 and 9). Then your team will have 5-6 minutes to answer them as a team (numbers 108 and 109).

Your individual grade, out of ten points, includes .5 point for each correctly bubbled/unbubbled circle (minimum score is 1pt for being here).

8. Given a 3×3 matrix A and the matrix

$$B = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

which of the following are true? (You might want to do some scratch work!)

- (a) BA performs the row operation $2R_2 + R_1 \rightarrow R_2$
- (b) BA performs the row operation $2R_2 + R_1 \rightarrow R_1$
- (c) AB performs the row operation $2R_2 + R_1 \rightarrow R_2$
- (d) AB performs the row operation $2R_2 + R_1 \rightarrow R_1$
- (e) None of the above.

8. For the same matrix B as above, which are true?

- (a) B can perform a row operation on A
- (b) B can perform a column operation on A
- (c) B is singular
- (d) B is nonsingular and

$$B^{-1} = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- (e) None of the above.